

PROFESSIONAL EXPERIENCE

- Research Lead**
Tutor Intelligence
• Foundation model research and training
- June 2026 — Present
Watertown, MA
- Advisor**
Tola Capital
• Technical advisor on robotics and AI investments
- May 2026 — Present
Seattle, WA
- Principal Research Lead**
Research Scientist
RAI Institute (formerly Boston Dynamics AI Institute)
- September 2023 — June 2026
Dec 2022 — August 2023
Cambridge, MA
- Pioneered handheld force-based data collection system for robot training, capturing force, vision, and proprioception for in-the-wild demonstrations
- Built video prediction system for robot policies using internet-scale video as prior (early 2023, before the wave); requires only limited real-world demonstrations
- Designed multimodal architectures combining novel sensors with internet-scale priors for improved behavioral execution (publication pending)
- Developed gradient-free RL enabling online learning with non-differentiable semantic functions (U.S. Patent pending, May 2025)
- Created task definition framework that makes human demonstrations more learnable for robots
- Built ROS 2 interfaces and controllers for custom grippers leveraging novel force/torque sensors
- Founded two research teams to scale innovations: Foundation Models (10+ researchers) and Capture (30 people)
- Built 320 H100 GPU infrastructure for large-scale training; created institute-wide policy execution framework
- Published at CoRL 2024, IJCAI 2024, ICML 2023; co-authoring “Elephants Don’t Write Sonnets” (MIT Press 2026)
- Established research partnerships with Google, Columbia, ETH Zurich, Agile Robots
- Staff Software Engineer**
Righthand Robotics
• Worked on vision and planning team to improve pick and place performance for Rightpick systems
• Position ended due to company layoffs
- August 2022 — October 2022
Somerville, MA
- CEO / Co-Founder, Odefi Inc.**
Columbia IBM Blockchain Accelerator
• Invented liquidity automation system for MakerDAO network; architected smart contract infrastructure
• Led technical architecture, product development, and investor pitches as part of Columbia IBM Blockchain Accelerator
- Mar 2019 — Jan 2022
New York, NY
- Research Fellow**
Army Research Lab
• Won 1st place MineRL BASALT Competition (NeurIPS 2021): invented hierarchical agent combining learned and engineered components
• Invented mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist)
• Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)
• Built simulation-to-real transfer pipelines for robotic grasping research
- Sep 2018 — July 2022
Aberdeen, MD
- Engineering Intern**
Goldman Sachs — Margin Technology
• Built graph-based calculation prioritization system (Neo4j) optimizing ordering across 10K+ daily margin computations
• Developed Java/Angular.js system for dynamic prioritization weight adjustment
- Jun 2016 — Aug 2016
New York, NY
- Engineering Intern**
Goldman Sachs — Valuations Technology
• Rebuilt FVA Gating Tool for derivatives valuations; implemented high-performance grid (100K+ rows, sub-second rendering)
- May 2015 — Aug 2015
New York, NY
- Engineering Intern**
Streakfire LLC
• Produced ad campaign in Puerto Rico to promote technical accelerator for job creation
• Coordinated with local government to leverage publicity expertise and infrastructure
- Jun 2014 — Sep 2014
Wayland, MA and Dorado, PR

TECHNICAL EXPERIENCE

Multiple View Performers for Shape Completion

Dec 2021 — Present

Robotics at Google, Army Research Lab, Columbia University

New York, NY

- Researched novel deep learning approach for multiple view completion without registering views
- Developed a process to leverage Performer attention layers developed by Google to encode multiple reconstruction images
- Work submitted to ICRA 2023

Data Driven Strand Simulation

Dec 2021 — Present

Columbia University

New York, NY

- Researched hair simulation algorithms to run in graph neural networks offering a 400% speedup
- Developed methods to compare python vs. C++ pytorch models
- In submission to TOG (ACM Transaction on Graphics)

MineRL Basalt Competition

Jul 2021 — Dec 2021

Neurips 2021

New York, NY

- Researched the intersection of engineered and learned knowledge to develop an autonomous Minecraft agent using human demonstration data and won first place at MineRL Basalt at Neurips 2021 in collaboration with ARL and UMBC
- Researched learned visual navigation methods and developed a CNN state classifier using human-labeled data
- Work published at AAAI-Make 2022 and presented at Neurips 2021

Mobile Manipulation Leveraging Multiple Views

Jan 2020 — Oct 2022

Columbia Robotics Lab

New York, NY

- Researched deep learning approaches to mobile manipulation without localizing the robot at runtime
- Explored novel simulation based techniques for generating data with real-world scanned environments
- Improved previous navigation work via predicted panoramic target goals from nearby environment reconstruction
- Published to IROS 2022 and nominated for best paper in mobile manipulation

Learning from Electromyography Synergies to Grasp Novel Objects by Superquadric Representation

Jun 2018 — May 2020

Columbia Robotics Lab

New York, NY

- Collaborated with students to create a system for learning grasp synergies using a CTRL-Labs arm band and mapping the EMG signals to an anthropomorphic Seed robotic hand
- Helped develop an algorithm for planning and executing grasps using superquadric representations of novel objects with successful real-world demonstrations
- Work presented at Columbia Data Science Day 2019

Learning Your Way Without Map or Compass: Panoramic Target Driven Visual Navigation

Jan 2018 — Sep 2019

Columbia Robotics Lab

New York, NY

- Researched novel visual navigation methodology using RGBD panoramic target goals and behavioral cloning
- Developed a system architecture to embed images using an autoencoder and a policy model to control the robot
- Explored optimization strategies to develop training data from real-world environments without human intervention
- Work published to IROS 2020 and presented at NERC 2019

Multi-Modal Geometric Learning for Grasping and Manipulation

Sep 2017 — Sep 2019

Columbia Robotics Lab

New York, NY

- Incorporated tactile information to estimate shape geometry using vision and touch via deep learning
- Created a novel machine learned model for estimating shape via multi-modal information
- Work published to ICRA 2019 and to a RSS 2017 workshop

Human Robot Interface for Assistive Grasping

Jan 2017 — Dec 2018

Columbia Robotics Lab and CUMC

New York, NY

- Created a novel interface for enabling robot control for spinal cord injury patients using an sEMG device
- Benchmarked the sEMG interface against other modalities including an Amazon Echo and a toggle switch
- Work presented at Columbia Data Science Day 2018

Research in Bit Width Resolution

Jan 2016 — Dec 2016

Columbia University

New York, NY

- Worked with Professor Stephen Edwards at Columbia University to add Z3's SMT framework to an existing compiler project in order to resolve variable bit widths at compile time

Research in Shape Completion

Sep 2016 — May 2017

Columbia Robotics Lab

New York, NY

- Worked with Professor Peter Allen in the Columbia Robotics Lab at Columbia University to optimize an existing platform utilizing CUDA
- Evaluated the ability to utilize a semantic pre-processor to identify objects in a scene to be completed using the existing tool

Research in Data Visualization

Columbia University

Sep 2015 — Dec 2015

New York, NY

- Worked with Professor John Kender at Columbia University to provide visualization of the correlation between visual and textual memes in online video data and provided an analysis on the most effective ways of visualizing co-clustered data

Research in the Production of Litecoin ASICs

Columbia University

Jan 2014 — May 2014

New York, NY

- Cooperated with Professor Simha Sethumadhavan on the feasibility of producing Litecoin Mining ASICs
- Independently designed all of the ASIC schematics, performed cost-benefit analysis of the ASIC and maintained knowledge on which crypto-currencies were most profitable at any time

Written Values Affirmation Intervention to Identify the Unique Linguistic Features of Stigmatized Groups

LIRSM

Sep 2014 — Jan 2015

New York, NY

- Responsible for developing web application in Node.js and Mongo to easily add and retrieve information from csv files
- Participated in IT work and assisted with sessions in informing individuals on using UNIX
- Developed strategies to acquire study data more efficiently and help audit costs on services

Walking in Their Shoes: Poverty in America

Yale

Sep 2014 — Jan 2015

New Haven, CT

- Developed interface and game for flexible assignment of agency to test subjects to explore the impact that games have on depictions of poverty
- Worked with MongoDB and Angular.js to create flexible tiled gameplay

PUBLICATIONS

1. Surendran, Vidullan, Neehar Peri, and **David Watkins**. "Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts." Conference on Robot Learning (CoRL), 2026 (pending acceptance). <https://nperi-rai.github.io/bridge-project/>
2. Kapoor, Kunal, Shiraz Khan, Joseph McCalmon, Jesse Michel, Arif Mohammed, Katerina Nikiforova, Adam Oppenheimer, Victor Szabo, and **David Watkins**. "Maintaining Demonstration Quality in a 100-Robot Teleoperation Pipeline." Workshop on "It's the Demos: A Deep Look at the Role of Demonstration Quality in Imitation-Based Robot Manipulation," Robotics: Science and Systems (RSS), Sydney, Australia, 2026.
3. Tellex, Stefanie, and **David Watkins**. "Elephants Don't Write Sonnets: The Grounded Turing Test for Embodied AI." 2025. <https://h2r.cs.brown.edu/wp-content/uploads/tellexwatkins2026.pdf>
4. Shang, Jinghuan, Karl Schmeckpeper, Brandon B. May, Maria Vittoria Minniti, Tarik Kelestemur, **David Watkins**, and Laura Herlant. 'Theia: Distilling Diverse Vision Foundation Models for Robot Learning'. In 8th Annual Conference on Robot Learning, 2024. <https://openreview.net/forum?id=yLZHvUcl>.
5. Cohen, Vanya, Jason Xinyu Liu, Raymond Mooney, Stefanie Tellex, and **David Watkins**. "A Survey of Robotic Language Grounding: Tradeoffs Between Symbols and Embeddings." arXiv preprint arXiv:2405.13245 (2024).
6. Novoseller, Ellen, Vinicius G. Goecks, **David Watkins**, Josh Miller, and Nicholas Waytowich. "DIP-RL: Demonstration-Inferred Preference Learning in Minecraft." arXiv preprint arXiv:2307.12158 (2023).
7. Milani, S., Kanervisto, A., Ramanauskas, K., Schulhoff, S., Houghton, B., Mohanty, S., ... & Shah, R. (2023). Towards Solving Fuzzy Tasks with Human Feedback: A Retrospective of the MineRL BASALT 2022 Competition. arXiv preprint arXiv:2303.13512.
8. Choromanski, K.M., Sehanobish, A., Lin, H., Zhao, Y., Berger, E., Parshakova, T., Pan, A., **Watkins, D.**, Zhang, T., Likhoshesterov, V. and Chowdhury, S.B.R., 2023, July. Efficient graph field integrators meet point clouds. In International Conference on Machine Learning (pp. 5978-6004). PMLR.
9. Hu, Jiaheng, **David Watkins**, and Peter Allen. "Teleoperated Robot Grasping in Virtual Reality Spaces." arXiv preprint arXiv:2301.13064 (2023).
10. **Watkins-Valls, D.**, Allen, P., Choromanski, K., Varley, J., & Waytowich, N. (2022). Multiple View Performers for Shape Completion. arXiv preprint arXiv:2209.06291.
11. **Watkins, David Joseph**. (2022). Learning Mobile Manipulation. Columbia University. <https://doi.org/10.7916/V9YM-TQ84>
12. **Watkins-Valls, D.**, Maia H., Varley J., Seshadri M., Sanabria J., Waytowich, N., & Allen, P. (2022). Mobile Manipulation Leveraging Multiple Views. 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2022
13. Goecks, Vinicius G., et al. "Combining Learning from Human Feedback and Knowledge Engineering to Solve Hierarchical Tasks in Minecraft." ArXiv:2112.03482 [Cs], Dec. 2021. arXiv.org, <http://arxiv.org/abs/2112.03482>. Accepted to AAAI-Make 2022.

14. **Watkins-Valls, D.**, Xu, J., Waytowich, N., & Allen, P. (2020). Learning your way without map or compass: Panoramic target driven visual navigation. 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2020
15. Akinola, Iretiayo, Zizhao Wang, Junyao Shi, Xiaomin He, Pawan Lapborisuth, Jingxi Xu, **David Watkins-Valls**, Paul Sajda, and Peter Allen. "Accelerated Robot Learning via Human Brain Signals." In 2020 IEEE International Conference on Robotics and Automation (ICRA), pp. 3799-3805. IEEE, 2020.
16. Wu, B., Akinola, I., Gupta, A., Xu, F., Varley, J., **Watkins-Valls, D.**, & Allen, P. K. (2020). Generative Attention Learning: a "GenerAL" framework for high-performance multi-fingered grasping in clutter. *Autonomous Robots*, 1-20.
17. **Watkins-Valls, D.**, Varley, J. & Allen, P. Multi-Modal Geometric Learning for Grasping and Manipulation. 2019 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2019.
18. Abhi Gupta, Jingya Bi, Ashwin Jayaraman, Max Xu, **David Watkins** and Professor Peter Allen. "Learning from Electromyography Synergies to Grasp Novel Objects by Super Quadric Representation (Poster)" In: Columbia Data Science Day (2019).
19. Jacob Varley, **David Watkins-Valls**, and Peter Allen. "Multi-Modal Geometric Learning for Grasping and Manipulation (Poster)". In: Columbia Data Science Day (2018).
20. Jacob Varley, **David Watkins**, and Peter Allen. "Visual-Tactile Geometric Reasoning (Abstract and Poster)". In: Data-Driven Manipulation workshop, Robotics: Science and Systems (2017).
21. **David Watkins-Valls**, Chaiwen Chou, Caroline Weinberg, Jacob Varley, Lynne Weber, Adam Blanchard, Peter Allen, Joel Stein "Human Robot Interface for Assistive Grasping (Poster)". In: New England Manipulation Symposium (2017).
22. **David Watkins-Valls** "Script Mining With ASICs" (2014).

WRITING

1. **Watkins, David**, and Stefanie Tellex. "Diffusion Policies are Fancy Lookup Tables." *What to Tell the Robot*, July 21, 2026. <https://whattotelltherobot.com/p/diffusion-policies-are-fancy-lookup>
2. Tellex, Stefanie, and **David Watkins**. "We Got a New Nine." *What to Tell the Robot*, July 11, 2026. <https://whattotelltherobot.com/p/we-got-a-new-nine>
3. Tellex, Stefanie, and **David Watkins**. "I Need Space." *What to Tell the Robot*, July 6, 2026. <https://whattotelltherobot.com/p/i-need-space>
4. Tellex, Stefanie, and **David Watkins**. "Why Robotics is a Pre-Paradigm Field." *What to Tell the Robot*, June 6, 2026. <https://whattotelltherobot.com/p/why-robotics-is-a-pre-paradigm-field>
5. Tellex, Stefanie, and **David Watkins**. "I Got Sycophanted." *What to Tell the Robot*, May 20, 2026. <https://whattotelltherobot.com/p/i-got-sycophanted>
6. Tellex, Stefanie, and **David Watkins**. "How to Build Safe AI (Without Making the AI Safe)." *What to Tell the Robot*, May 13, 2026. <https://whattotelltherobot.com/p/how-to-build-safe-ai-without-making>
7. **Watkins, David** (contributing author). "Getting a Grip on Robotic Data Collection." *RAI Institute Blog*, May 12, 2026. <https://rai-inst.com/resources/blog/handheld-robotic-data-collection/>
8. Tellex, Stefanie, and **David Watkins**. "The First Hit is Free." *What to Tell the Robot*, May 7, 2026. <https://whattotelltherobot.com/p/the-first-hit-is-free>
9. **Watkins, David**, and Stefanie Tellex. "What Babies Know That Robots Don't." *What to Tell the Robot*, April 15, 2026. <https://whattotelltherobot.com/p/what-babies-know-that-robots-dont>
10. **Watkins, David**, and Stefanie Tellex. "Consciousoids." *What to Tell the Robot*, March 30, 2026. <https://whattotelltherobot.com/p/consciousoids>
11. Tellex, Stefanie, and **David Watkins**. "Elephants Still Don't Play Chess." *What to Tell the Robot*, March 2, 2026. <https://whattotelltherobot.com/p/elephants-still-dont-play-chess>
12. Tellex, Stefanie, and **David Watkins**. "Radiate Love: a 2025 Retrospective." *What to Tell the Robot*, February 3, 2026. <https://whattotelltherobot.com/p/radiate-love-a-2025-retrospective>
13. **Watkins, David**, and Stefanie Tellex. "Satisfying a 31-Year-Old Karaoke Dream." *What to Tell the Robot*, January 16, 2026. <https://whattotelltherobot.com/p/satisfying-a-31-year-old-karaoke>

14. Tellex, Stefanie, and **David Watkins**. "Where the Curves Cross." *What to Tell the Robot*, January 9, 2026.
<https://whattotelltherobot.com/p/where-the-curves-cross>
15. **Watkins, David**. "2025 in Review: The Year of Showing Up." *What to Tell the Robot*, December 31, 2025.
<https://whattotelltherobot.com/p/2025-in-review-the-year-of-showing>
16. **Watkins, David**, and Stefanie Tellex. "The Deer Island Marvel." *What to Tell the Robot*, December 5, 2025.
<https://whattotelltherobot.com/p/the-deer-island-marvel>
17. Tellex, Stefanie, and **David Watkins**. "Qualitative Simulation of Swine Production." *What to Tell the Robot*, November 19, 2025. <https://whattotelltherobot.com/p/qualitative-simulation-of-swine-production>
18. Tellex, Stefanie, and **David Watkins**. "The Grounded Turing Test." *What to Tell the Robot*, September 2, 2025.
<https://whattotelltherobot.com/p/the-grounded-turing-test>
19. **Watkins, David**, and Stefanie Tellex. "From Physics to Embodied Intelligence." *What to Tell the Robot*, August 27, 2025.
<https://whattotelltherobot.com/p/from-physics-to-embodied-intelligence>
20. Tellex, Stefanie, and **David Watkins**. "Taking a Quadruped Robot to School." *What to Tell the Robot*, August 20, 2025.
<https://whattotelltherobot.com/p/taking-a-quadruped-robot-to-school>
21. **Watkins, David**, and Stefanie Tellex. "Elephants Don't Write Sonnets." *What to Tell the Robot*, August 6, 2025.
<https://whattotelltherobot.com/p/elephants-dont-write-sonnets>
22. Tellex, Stefanie, and **David Watkins**. "What to Tell the Robot." *What to Tell the Robot*, July 18, 2025.
<https://whattotelltherobot.com/p/what-to-tell-the-robot>

EDUCATION

PhD in Computer Science , <i>Columbia University</i>	Sep 2017 — May 2022
Advisor: <i>Prof. Peter Allen</i> , Thesis: <i>Learning Mobile Manipulation</i>	
<i>Army Research Lab Research Fellow</i>	Sep 2018 — July 2022
MPhil in Computer Science , <i>Columbia University</i>	Sep 2017 — May 2019
MS in Computer Science , <i>Columbia University</i> , 4.0 GPA	Sep 2016 — May 2017
<i>CA Fellowship</i>	Sep 2016 — Jan 2017
BS in Computer Science , <i>Columbia University</i> , 3.7 GPA	Sep 2012 — May 2016
Marian High School	Oct 2010 — May 2012
<i>Class President</i>	
<i>Salutatorian</i>	
<i>National Honors Society</i>	

AWARDS

Best Paper in Mobile Manipulation Finalist , <i>Kyoto, Japan</i>	Oct 2022
<i>Selected for my work titled Mobile Manipulation Leveraging Multiple Views submitted to IROS 2022</i>	
International House Member , <i>New York, NY</i>	Sep 2016 — May 2018
<i>Competitively selected scholars and young professionals from around the world who are challenged to become globally-minded leaders</i>	
CA Fellowship , <i>Columbia University</i>	Sep 2016 — Dec 2016
<i>MS students who have proven themselves to be exceptional will receive paid tuition and stipend</i>	
Goldman Sachs Code Golf Champion , <i>New York, NY</i>	Aug 2016
<i>Awarded for shortest possible source code that implements a certain algorithm</i>	
Dean's List , <i>Columbia University</i>	Spring 2013, Spring 2015, Spring 2016
<i>A list of students recognized for academic achievement during a semester by the dean of the college they attend</i>	
Residential Incubator Fellow , <i>Columbia University</i>	Sep 2012 — May 2014
<i>Students interested in entrepreneurship participating in a student incubator</i>	

ACTIVITIES

Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer, RSS	2026
New England Manipulation Symposium (NEMS) Co-Organizer	2025
IJRR Reviewer	2023
NSF Reviewer	2023
ICRA Paper Reviewer	2022, 2025
Computer Science Student Faculty Representative	2018 — 2020

SKILLS

Languages	Python, C++, CUDA, ROS 2, PyTorch, Bash, $\text{\LaTeX}$, SQL
ML/AI	Foundation Models, Transformers, Diffusion Models, Policy Learning, Distributed Training
Infrastructure	Kubernetes, Docker, H100/A100 Clusters, Slurm, Ray, Git, CI/CD
Robotics	Manipulation, Navigation, SLAM, Simulation (PyBullet, MuJoCo, Isaac), ROS 2
Leadership	Team Building (0→13+ FTEs), Hiring (200+ interviews), Strategic Partnerships
Communication	English, Spanish

VOLUNTEER WORK

Multisensory Reading Centers of Puerto Rico , San Juan, PR	Sep 2017 — Present
<i>Over 150 hours of volunteer service by performing IT help to provide access to effective literacy instruction for struggling readers</i>	
Watkins-Valls Family Foundation , Boston, MA	Sep 2013 — Present
<i>Providing scholarships and academic support to underprivileged students in Massachusetts, New York, and Puerto Rico</i>	
Pine St. Inn , Boston, MA	Dec 2010 — May 2012
<i>Over 25 hours of volunteer service at a soup kitchen for the homeless</i>	
Sisters of St. Joseph , Cambridge, MA	Dec 2010 — May 2012
<i>Over 25 hours of volunteer service through entertaining and assisting retired nuns</i>	

PRESENTATIONS

What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels , Extropic, Waltham, MA	Jul 2026
<i>Invited talk on robot data collection, from co-designed handheld force-based capture to fleet-scale data flywheels, presented to the Extropic team</i>	
Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer , RSS 2026, Sydney	Jul 2026
<i>Co-organized workshop on the role of tactile sensing in robotic foundation models and large-scale data collection</i>	
What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels , RSS 2026 Workshop (It's the Demos), Sydney	Jul 2026
<i>Invited workshop talk tracing robot data collection from co-designed handheld force-based capture to fleet-scale data flywheels, drawing on work at the RAI Institute and Data Factory 1 (DF1)</i>	
Why Robots? Data-Driven Manipulation for the World's Hardest Sorting Problem , Brown University	Feb 2026
<i>Guest lecture in Stefanie Tellex's class on robotic manipulation for landfill restoration, covering multimodal sensing, force-based data collection, and the capability gap between warehouse and unstructured-environment robots</i>	
New England Manipulation Symposium (NEMS) 2025 Co-Organizer , MIT	Apr 2025
<i>Co-organized conference with Lael Odhner and Kaitlyn Becker; coordinated paper acceptances, speaker scheduling, and venue logistics</i>	
The Future of Intelligent Robotics , Dr. Waku YouTube Interview	2025
<i>Featured interview discussing robotics' "ChatGPT moment" and the future of embodied AI</i>	
Lions in AI Panel , Columbia Alumni Association of Boston	2025
<i>Panel discussion on AI and robotics for Columbia alumni community</i>	
GTC 2025 Summary , RAI Institute	Mar 2025
<i>Presented key takeaways from NVIDIA GPU Technology Conference to institute leadership</i>	
Theia: Distilling Diverse Vision Foundation Models , CoRL 2024, Munich	Nov 2024
<i>Published work on foundation model distillation for robot learning at Conference on Robot Learning</i>	
Foundation Models for Robotics , Duke University	Feb 2024
<i>Presentation on state of the art foundation models for robotic learning</i>	
Helios: High Dimensional Predictive Foundation Models as Knowledge Transfer Systems , Brown University	Jul 2023
<i>Building large scale video predictive foundation models for robotics at the AI Institute</i>	
From One to Many: How to leverage multiple views in shape completion , Robotics at Google	Dec 2022
<i>A discussion on how to leverage multiple views for shape completion presented to researchers at Robotics at Google</i>	
Learning Mobile Manipulation , Harvard - Harvard Biorobotics Laboratory	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Robert Howe's lab</i>	
Learning Mobile Manipulation , Tufts - Human Robotics Interaction Laboratory	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Matthias Scheutz's lab</i>	
Learning Mobile Manipulation , MIT - Improbable AI Lab	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Pulkit Agrawal's lab</i>	
Learning Mobile Manipulation , Boston Dynamics	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	

Learning Mobile Manipulation , Boston Dynamics AI Institute <i>Presented a collection of works done during my Ph.D.</i>	Nov 2022
Learning Mobile Manipulation , MITRE <i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	Nov 2022
Learning Mobile Manipulation , Rotor.ai <i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	Oct 2022
Learning Mobile Manipulation , STR <i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	Oct 2022
Mobile Manipulation Leveraging Multiple Views , IROS 2022 <i>Presented work on localization free mobile manipulation that was nominated for best paper in mobile manipulation</i>	Oct 2022
P.h.D. Defense: Learning Mobile Manipulation , Columbia University <i>Presented my dissertation defense enabling mobile robots to manipulate objects</i>	May 2022
Minecraft Basalt Interview , Yannic Kilcher YT <i>Interviewed on recent first place win in Minecraft Basalt RL competition.</i>	Jan 2022
Learning Mobile Manipulation , ROAM Lab <i>Presented work done in preparation of defense of my thesis to the ROAM lab at Columbia University.</i>	Dec 2021
Learning Mobile Manipulation , CAIR Lab <i>Presented my thesis proposal to the CAIR lab at Columbia University.</i>	Feb 2021
IROS 2020: Learning Your Way Without Map or Compass , IROS 2020 <i>Presented my Learning Your Way work at IROS 2020 via the online conference.</i>	Oct 2020
Demystifying the Dissertation , Columbia University <i>Presented my thesis proposal and help current graduate students understand the process of completing a PhD.</i>	Jun 2020
Thesis Proposal: Learning Mobile Manipulation , New York, NY <i>Presented and defended my thesis proposal in learning mobile manipulation in January 2020.</i>	Jan 2020
Research Talk , Harlem Children's Zone STEM Exposure <i>Shared my research as part of an initiative to help teach children in Harlem methods in STEM research.</i>	Nov 2019
Learning Your Way Talk , NYU Reading Group <i>Presented Learning Your Way work to the NYU robotics reading group.</i>	Oct 2019
Learning Your Way Talk , NERC 2019 <i>Presented Learning Your Way work in front of audience of robotics researchers at NERC 2019 conference.</i>	Oct 2019
Visual Tactile Completion , Emptor Lightning Talks <i>Presented work on visual tactile grasping as well as next steps as part of Emptor's lightning talks.</i>	Sep 2019
Odefi Pitch , Columbia IBM Blockchain Accelerator Demo Day 2019 <i>Enabling credit default swaps on the ethereum network pitched to investors at the capstone event for the Columbia IBM Blockchain Accelerator.</i>	May 2019
Visual Tactile Grasping , Samsung Research NYC <i>Presented work on visual tactile grasping as well as next steps.</i>	July 2019
Candidacy Exam: Simulation for Real World Robotics , New York, NY <i>A high-level overview of how real-world robotics can be enabled through simulation.</i>	May 2019
Using Simulation to Enable Generated Art and Robotics , Making Art in the Age of Algorithms <i>A high-level overview of how robotics can be enabled through simulation as part of a series of lightning talks about art and algorithms.</i>	Dec 2018
Visual Tactile Completion Poster , Data Science Day 2018 <i>This work provides an architecture that incorporates depth and tactile information to create rich and accurate 3D models useful for robotic manipulation tasks presented at Data Science Day 2018 at Columbia University.</i>	Mar 2018
Lecture: ROS Tutorial , New York, NY <i>An introductory tutorial on ROS and use of robotics in the Columbia Robotics Lab to aspiring roboticists.</i>	Jan 2018
Providing Context to Startup Culture , New York, NY <i>An analysis on the effectiveness of a startup based on the type of culture it maintains as well as effects on profit/loss.</i>	May 2016

MEMBERSHIPS

ACM	2016 — Present
IEEE	2016 — Present
SHPE	2018 — Present
AAAI	2022 — Present

TEACHING EXPERIENCE

Humanoid Robotics (COMSW 6731) , Teaching Assistant, Graduate Level	Spring 2018
Computational Aspects of Robotics (COMSW 4733) , Teaching Assistant, Graduate Level	Fall 2017
Programming Languages and Translators (COMSW 4115) , Teaching Assistant, Graduate Level	Fall 2016
Object Oriented Programming and Design in Java (COMS 1007) , Teaching Assistant, Undergraduate Level	Fall 2016

Programming Languages and Translators (COMSW 4115) , Teaching Assistant, Graduate Level	Spring 2016
Object Oriented Programming and Design in Java (COMS1007) , Teaching Assistant, Undergraduate Level	Fall 2014
Fundamentals of Computer Systems (CSEE3827) , Teaching Assistant, Undergraduate Level	Spring 2014
Object Oriented Programming and Design in Java (COMS1007) , Teaching Assistant, Undergraduate Level	Fall 2013

REFERENCES

References available upon request.